

DNA/Protein Sequence Analysis Using BLAST

CodeAlpha Bioinformatics Internship — Task 1

1. Objective

To obtain a protein sequence from UniProt and analyze it using NCBI Protein BLAST in order to identify homologous sequences and document sequence identity, query coverage, alignment score, E-value and pairwise alignment characteristics.

2. Introduction

Bioinformatics combines biological information with computational methods to analyze biological sequences. BLAST (Basic Local Alignment Search Tool) is used to compare a query nucleotide or protein sequence against sequences in a database and identify similar or homologous sequences. The results can be interpreted using measures such as percentage identity, query coverage, alignment score and E-value.

3. Protein Selected

Parameter	Observed value
Protein	Hemoglobin subunit beta (HBB)
Organism	Homo sapiens
UniProt accession	P68871
Molecule type	Amino acid / protein
Sequence length	147 amino acids
Sequence status	Complete

4. Methodology

The HBB protein entry P68871 was accessed in UniProt and its complete 147-amino-acid sequence was copied in FASTA form. The sequence was submitted to NCBI Protein BLAST (BLASTP) using the ClusteredNR database. Significant matches were examined in the BLAST results and the pairwise alignment view was used to inspect identities, positives, gaps, score and E-value.

Figure 1. UniProt sequence record for HBB_HUMAN (P68871)

Function	P68871 · HBB_HUMAN HBB	Tools · Download · Add · Community curated (3) · Add a publication · Entry feedback
Names & Taxonomy	Entry · Variant viewer 817 · Feature viewer · Genomic coordinates · Publications · External links · History	
Subcellular Location	Sequence ⁱ	
Disease & Variants	Sequence status ⁱ Complete	
PTM/Processing	See also sequence in UniParc or sequence clusters in UniRef	
Expression	Tools · Download · Add · Highlight · Copy sequence	
Interaction	Length 147	Last updated 2007-01-23 v2
Structure	Mass (Da) 15,998	MD5 Checksum ⁱ 209D686939D0B8D1EA089368150140E2
Family & Domains	MVHLTPEEKS ¹⁰ AVTALMGKVN ²⁰ VDEVGGEALG ³⁰ RLLVVPWTQ ⁴⁰ RFFESFGDLS ⁵⁰ TPDVAMGNPK ⁶⁰ VKAHGKKVLG ⁷⁰ AFSQGLAHL ⁸⁰ NLKGTFTALS ⁹⁰ ELHCDKLHVD ¹⁰⁰ PENFRLLGNV ¹¹⁰ LVCVLAHFG ¹²⁰ KEFTPPVQAA ¹³⁰	
Sequence	YQKVVGAVN ¹⁴⁰ ALAHKYH	
Similar Proteins		Help

Source: UniProt record viewed during the analysis.

5. BLAST Search Results

The BLAST search produced multiple significant sequence matches. The visible top hits all showed 100% query coverage in the cluster summary. The principal observed values are summarized below.

BLAST hit	Query cover	% identity	E-value	Max score
Hb Monza protein [H. sapiens]	100%	86.47%	2e-97	288
beta-globin [H. sapiens]	100%	93.20%	2e-93	277
Hb beta-like [C. lupus familiaris]	100%	89.80%	1e-92	275
Hb beta-like [A. melanoleuca]	100%	89.80%	7e-92	273
Hb beta [R. naso]	100%	89.12%	2e-91	271

Interpretation of the BLAST parameters

- **Query coverage:** 100% for the visible top cluster hits indicates that the full 147-residue query was covered by the reported alignment.
- **Percentage identity:** The visible hits showed approximately 86–93% identity. For example, the beta-globin hit showed 137 identical residues out of 147 aligned positions (93%).
- **E-value:** The observed E-values were extremely small (for example, 2e-93 and 2e-97), indicating highly significant matches under the BLAST scoring model.
- **Max score:** The visible scores ranged from 271 to 288 bits among the listed top hits. A higher score represents a stronger alignment under the scoring system used.

Figure 2. NCBI BLAST cluster results

The screenshot displays the NCBI BLAST results page for the query RID-B1T9PC79014. The top section includes search parameters and a 'Filter Results' panel. The 'Filter Results' panel shows 'Organism' set to 'only top 20 will appear' and 'Percent Identity' set to '100%'. The 'Clusters' tab is selected, showing a table of clusters producing significant alignments. The table has columns for Cluster Composition, Cluster Ancestor, Cluster Representative Sequence, Max Score, Total Score, Query Cover, E value, Per. Ident, Acc. Len, and Accession. The table lists several clusters, including 'human', 'dog_coyote_wolf_fox', 'placentalis', 'bats', 'giant_panda', and 'large_flying_fox'. The 'human' cluster is highlighted, showing a representative sequence of 'Hb Monza protein [Homo sapiens]' with a Max Score of 288, Total Score of 288, Query Cover of 100%, E value of 2e-97, Per. Ident of 86.47%, Acc. Len of 170, and Accession of XEF57200.1.

Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
1 member(s), 1 organism(s)	human	Hb Monza protein [Homo sapiens]	288	288	100%	2e-97	86.47%	170	XEF57200.1
1 member(s), 1 organism(s)	human	beta-globin [Homo sapiens]	277	277	100%	2e-93	93.20%	147	AA035597.1
9 member(s), 7 organism(s)	dog_coyote_wolf_fox	hemoglobin subunit beta-like [Canis lupus familiaris]	275	275	100%	1e-92	89.80%	147	NP_001257812.1
184 member(s), 76 organism(s)	placentalis	hemoglobin subunit delta [Alluopoda melanoleuca]	273	273	100%	7e-92	89.80%	147	NP_001281814.1
16 member(s), 17 organism(s)	placentalis	hemoglobin subunit beta-like [Alluopoda melanoleuca]	273	273	100%	7e-92	89.80%	147	NP_001281812.1
11 member(s), 8 organism(s)	bats	hemoglobin subunit beta [Rhinonycteris naso]	271	271	100%	2e-91	89.12%	147	KAM0815943.1
1 member(s), 1 organism(s)	giant_panda	hypothetical protein PANDA_015769 [Alluopoda melanoleuca]	275	446	100%	3e-91	90.48%	258	EFB18430.1
1 member(s), 1 organism(s)	large_flying_fox	TPA_globin A2 [Pteropus vampyrus]	272	272	100%	8e-91	88.44%	193	SAH82279.1

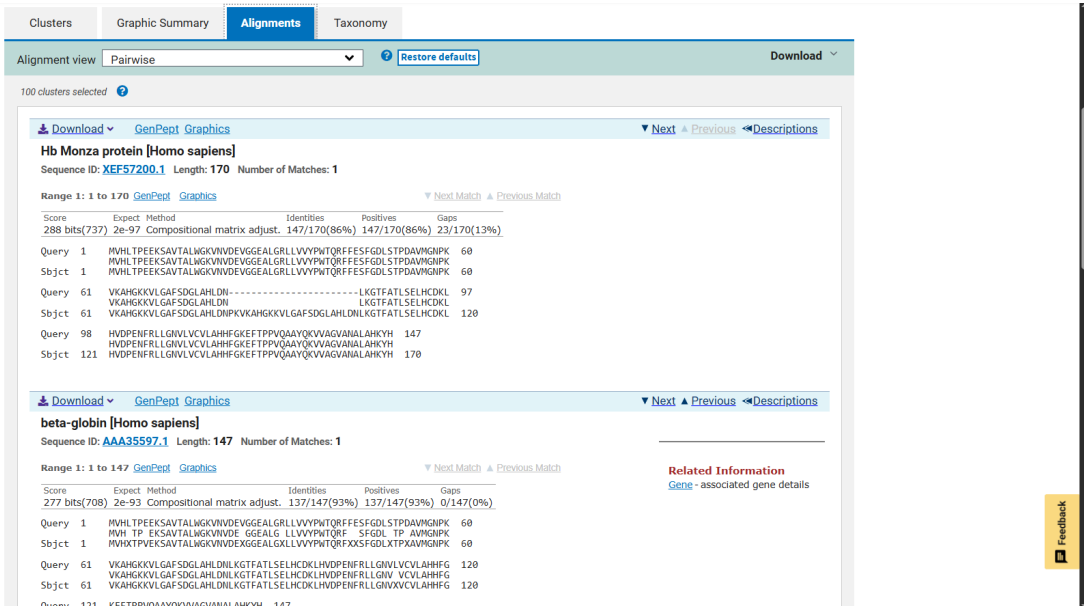
Source: NCBI BLAST results generated from the 147-aa HBB query.

6. Pairwise Alignment Analysis

The pairwise alignment view provided detailed alignment statistics for individual BLAST hits. For the beta-globin [Homo sapiens] hit (AAA35597.1), the alignment covered the complete 147-residue query and showed 137/147 identical positions, corresponding to 93% identity. The alignment had 0/147 gaps. The reported score was 277 bits (708) and the E-value was 2e-93.

Parameter	beta-globin [Homo sapiens] — AAA35597.1
Subject length	147 aa
Score	277 bits (708)
E-value	2e-93
Identities	137/147 (93%)
Positives	137/147 (93%)
Gaps	0/147 (0%)

Figure 3. Pairwise alignment results



Source: NCBI BLAST pairwise alignment view.

7. Discussion

The BLAST analysis identified several proteins with strong sequence similarity to the human HBB query. The beta-globin hit was 147 amino acids long, matching the query length, and had 93% sequence identity with no gaps in the reported alignment. The Hb Monza protein hit showed 86% identity over the reported 147/170 aligned positions and contained gaps because the subject sequence was longer. Together, these results demonstrate that the HBB query has strong similarity to related hemoglobin/globin proteins.

8. Conclusion

A complete 147-amino-acid human hemoglobin beta subunit sequence (UniProt P68871) was successfully analyzed using NCBI Protein BLAST. Multiple significant homologous sequences were identified. The observed high percentage identities, complete query coverage in the top cluster results and very small E-values support strong sequence similarity between HBB and the reported globin proteins. Pairwise alignment further demonstrated 93% identity and 0% gaps for the selected beta-globin hit.

9. References

- UniProt. HBB_HUMAN, accession P68871. UniProt Knowledgebase.
- NCBI. BLAST: Basic Local Alignment Search Tool. National Center for Biotechnology Information.

Note: Numerical results in this report are transcribed from the BLAST and UniProt screenshots generated during this analysis.